27-09-2024

NUMERICAL DIFFERENTIATION & INTEGRATION

- So far, we were finding the polynomial curve $y=f(x)$, passing through the $(n+1)$ ordered pairs (x_i, y_i), $i=0,1,2,\ldots\ldots,n$.
- Now we are trying to find the derivative values of such curves at a given $x=x_k$ (say), whose $x_0 < x_k < x_n$ (or even outside ~~et~~ the range but closer to starting or end ~~points~~ values).
- To get the derivative, we first find the curve $y=f(x)$ through the points and then differentiate and get its value at the required point.
- If the values of x are equally spaced, we get the interpolating polynomial due to Newton-Gregory. If the derivatives is required at a point nearer to the starting value in the table, we use Newton's forward interpolation formula.
- If we require the derivative at the end of the table, we use Newton's backward interpolation formula.
- If the value of the derivative is required near the middle of the table value we use one of the central difference interpolation formulae. In case of unequal intervals, we can use Newton's divided difference formula or Lagrange's interpolation formula to get the derivative value.

NEWTON'S FORWARD DIFFERENCE FORMULA TO GET THE DERIVATIVE

We have $n+1$ ordered pairs (x_i, y_i), i varies from 0 to n, $0 \le i \le n$. We have to find the derivative of $y=f(x)$ passing through the $n+1$ points at a point nearer to the starting value $x = x_0$.

Newton's forward difference interpolation formula is

$$y(x+uh) = y_u = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \cdots\cdots \quad -(1)$$

where $y(x)$ is a polynomial of degree 'n' in x, and

$$u = \frac{x - x_0}{h}.$$

Differentiating $y(x)$ with respect to x,

$$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}$$

$u = \frac{x-x_0}{h}$

$\therefore \frac{du}{dx} = \frac{1}{h}$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{h}\cdot\frac{dy}{du}$$

$u(u^2 - 3x + 2)$
$= u^3 - 3u^2 + 2u$
$(u^3 - 3u^2 + 2u)(u-3)$
$= u^4 - 3u^3 - 3u^3 + 9u^2 + 2u^2 - 6u$
$= u^4 - 6u^3 + 11u^2 - 6u$

$$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 + \frac{2u-1}{2!}\Delta^2 y_0 + \frac{3u^2 - 6u + 2}{3!}\Delta^3 y_0 + \frac{4u^3 - 18u^2 + 22u - 6}{24}\Delta^4 y_0 + \cdots\cdots\right] \quad -(2)$$

(2) gives the value of $\frac{dy}{dx}$ at general x, which may be anywhere in the interval.

In particular, at $x = x_0$, $u = \frac{x - x_0}{h} = 0$, then (2) reduces to

$$\left(\frac{dy}{dx}\right)_{x=x_0} = \left(\frac{dy}{dx}\right)_{u=0} = \frac{1}{h}\left[\Delta y_0 + \frac{-1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \cdots\right] \quad -(3)$$

↳ Newton's formula at beginning of the table.

Differentiating ② again wrt x,

$$\frac{d^2y}{dx^2} = \frac{d}{du}\left(\frac{dy}{dx}\right)\cdot\frac{du}{dx} \qquad \text{(Chain rule)}$$

$$\frac{d^2y}{dx^2} = \frac{1}{h}\frac{d}{du}\left(\frac{dy}{dx}\right) \qquad \left(\frac{du}{dx}=\frac{1}{h}\right)$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[0 + \Delta y_0 + \frac{u-1}{1}\Delta^3 y_0 + \frac{6u^2-18u+11}{12}\Delta^4 y_0 + \dots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_0 + (u-1)\Delta^3 y_0 + \frac{6u^2-18u+11}{12}\Delta^4 y_0 + \dots\right] \quad -④$$

Similarly, differentiating ④ wrt x,

$$\frac{d^3y}{dx^3} = \frac{d}{du}\left(\frac{d^2y}{dx^2}\right)\frac{du}{dx}$$

$$\frac{d^3y}{dx^3} = \frac{1}{h^3}\left[\Delta^3 y_0 + \frac{12u-18}{12}\Delta^4 y_0 + \dots\right] \quad -⑤$$

Equation ④ and ⑤ give second and third derivative value at any x. Putting $x=x_0$, $u=0$, in ④ and ⑤ we get

$$\left(\frac{d^2y}{dx^2}\right)_{x=x_0} = \left(\frac{d^2y}{dx^2}\right)_{u=0} = \frac{1}{h^2}\left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 + \dots\right]$$

$$\left(\frac{d^2y}{dx^2}\right)_{x=x_0} = \left(\frac{d^2y}{dx^2}\right)_{u=0} = \frac{1}{h^2}\left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 - \frac{5}{6}\Delta^5 y_0 + \frac{137}{180}\Delta^6 + \dots\right]$$

$$\left(\frac{d^3y}{dx^3}\right)_{x=x_0} = \frac{1}{h^3}\left[\Delta^3 y_0 - \frac{3}{2}\Delta^4 y_0 + \dots\right] \quad -⑦$$

Equation ⑥ and ⑦ gives the values of II nd and III rd derivative at the starting value $x=x_0$.

$$1+\Delta = E = e^{hD}$$

Taking log on both sides,

$$hD = \log(1+\Delta) = \Delta - \frac{1}{2}\Delta^2 + \frac{1}{3}\Delta^3 - \cdots$$

$$D = \frac{1}{h}\left(\Delta - \frac{1}{2}\Delta^2 + \frac{1}{3}\Delta^3 - \cdots\right)$$

Differentiating we get,

$$D^2 = \frac{1}{h^2}\left[\Delta^2 - \Delta^3 + \frac{11}{12}\Delta^4 - \cdots\right]$$

$$D^3 = \frac{1}{h^3}\left[\Delta^3 - \frac{3}{2}\Delta^4 + \cdots\right]$$

Alternative eqns using operators:

$$D = \frac{1}{h}\left(\Delta - \frac{1}{2}\Delta^2 + \frac{1}{3}\Delta^3 - \cdots\right)$$

$$D^2 = \frac{1}{h^2}\left[\Delta^2 - \Delta^3 + \frac{11}{12}\Delta^4 - \cdots\right]$$

$$D^3 = \frac{1}{h^3}\left[\Delta^3 - \frac{3}{2}\Delta^4 + \cdots\right]$$

NEWTON'S BACKWARD DIFFERENCE FORMULA TO COMPUTE THE DERIVATIVE

Newton's backward difference interpolation formula,

$$y(x) = y(x_n + Vh) = y_n + V\nabla y_n + \frac{V(V+1)}{2!}\nabla^2 y_n + \frac{V(V+1)(V+2)}{3!}\nabla^3 y_n + \cdots \quad -(8)$$

where, $V = \frac{x - x_n}{h}$

Differentiating (8) w.r.t x,

$$\frac{dy}{dx} = \frac{dy}{dv}\cdot\frac{dv}{dx} \qquad \left(\frac{dv}{dx} = \frac{1}{h}\right)$$

$$V(V^2+3V+2) = V^3 + 3V^2 + 2V$$

$$\frac{dy}{dx} = \frac{1}{h}\frac{dy}{dv}$$

$$= \frac{1}{h}\left[\nabla y_n + \frac{2V+1}{2}\nabla^2 y_n + \frac{3V^2+6V+2}{6}\nabla^3 y_n + \cdots\right]$$

$$\therefore \frac{dy}{dx} = \frac{1}{h}\left[\nabla y_n + \frac{(2v+1)}{2}\nabla^2 y_n + \frac{(3v^2+6v+2)}{6}\nabla^3 y_n + \frac{4v^3+18v^2+22v+12}{24}\nabla^4 y_n + \cdots\right] \quad -(9)$$

similarly,

Differentiating (9) w.r.t x,

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\nabla^2 y_n + (v+1)\nabla^3 y_n + \frac{6v^2+18v+11}{12}\nabla^4 y_n + \cdots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\nabla^2 y_n + (v+1)\nabla^3 y_n + \frac{6v^2+18v+11}{12}\nabla^4 y_n + \cdots\right] \quad -(10)$$

Illy,

$$\frac{d^3y}{dx^3} = \frac{1}{h^3}\left[\nabla^3 y_n + \frac{12v+18}{12}\nabla^4 y_n + \cdots\right] \quad -(11)$$

Equation (9), (10) and (11) gives first, second and third derivative at any general x.

Putting, $x = x_n$, $v = 0$. From (9), (10) and (11) we get,

$$\left(\frac{dy}{dx}\right)_{x=x_n} = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \frac{1}{4}\nabla^4 y_n + \cdots\right]$$

$$\left(\frac{d^2y}{dx^2}\right)_{x=x_n} = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \cdots\right]$$

$$\left(\frac{d^3y}{dx^3}\right)_{x=x_n} = \frac{1}{h^3}\left[\nabla^3 y_n + \frac{3}{2}\nabla^4 y_n + \cdots\right]$$

Alternatively,

$$1 - \nabla = E^{-1} = e^{-hD}$$

$$\Rightarrow \quad -hD = \log(1-\nabla)$$

$$D = \frac{1}{h}\log(\nabla - 1)$$

$$D = \frac{1}{h}\left[\nabla + \frac{1}{2}\nabla^2 + \frac{1}{3}\nabla^3 + \frac{1}{4}\nabla^4 + \cdots\right]$$

(don't write this formula for exam)

$$D = \frac{1}{h}\left[\nabla + \frac{1}{2}\nabla^2 + \frac{1}{3}\nabla^3 + \frac{1}{4}\nabla^4 + \cdots\right]$$

$$D^2 = \frac{1}{h^2}\left[\nabla^2 + \nabla^3 + \frac{11}{12}\nabla^4 + \cdots\right]$$

$$D^3 = \frac{1}{h^3}\left[\nabla^3 + \frac{3}{2}\nabla^4 + \cdots\right]$$

Q: Find the first two derivatives of $x^{1/3}$, at $x = 50$ and $x = 56$, from the given table.

x	50	51	52	53	54	55	56
$y = x^{1/3}$	3.6840	3.7084	3.7325	3.7563	3.7798	3.8030	3.8259

ans:

x	$y = x^{1/3}$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	Δ^6
50	3.6840						
		0.0244					
51	3.7084		−0.0003				
		0.0241		0			
52	3.7325		−0.0003		0		
		0.0238		0		0	
53	3.7563		−0.0003		0		0
		0.0235		0		0	
54	3.7798		−0.0003		0		
		0.0232		0			
55	3.8030		−0.0003				
		0.0229					
56	3.8259						

$x_0 = 50$ $\quad x_n = 56$

$$u = \frac{x - x_0}{h} = \frac{50 - 50}{1} = 0$$

By Newton's forward interpolation formula,

$$\left(\frac{dy}{dx}\right)_{x=x_0=5} = \left(\frac{dy}{dx}\right)_{u=0} = \frac{1}{h}\left[\Delta y_0 + \frac{-1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \dots\right]$$

$$\left(\frac{dy}{dx}\right)_{x=50} = \left[0.0244 - \frac{1}{2} \times -0.0003 + \frac{1}{3} \times 0 - 0\right]$$

$$\left(\frac{dy}{dx}\right)_{x=50} = 0.02455$$

$$\left(\frac{d^2y}{dx^2}\right)_{x=x_0} = \frac{1}{h^2}\left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 + \dots\right]$$

$$\left(\frac{d^2y}{dx^2}\right)_{x=50} = \frac{1}{1^2}\left[-0.0003\right] = -0.0003$$

$v=0$, $x=x_n$.

∴ By Newton's backward ~~dif~~ interpolation formula,

$$\left(\frac{dy}{dx}\right)_{x=x_n} = \left(\frac{dy}{dx}\right)_{v=0} = \frac{1}{h}\left[\Delta y_n + \frac{1}{2}\Delta^2 y_n + \frac{1}{3}\Delta^3 y_n + \frac{1}{4}\Delta^4 y_n + \dots\right]$$

$(h=1)$

$$\left(\frac{dy}{dx}\right)_{x=56} = \left(0.0229 + \frac{1}{2} \times -0.0003 + 0 + 0 + \dots\right)$$

$$\left(\frac{dy}{dx}\right)_{x=56} = 0.02275$$

$$\left(\frac{d^2y}{dx^2}\right)_{x=x_n} = \frac{1}{h^2}\left[\Delta^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \dots\right]$$

$$= \left[-0.0003 + 0 + \dots\right]$$

$$\therefore \left(\frac{d^2y}{dx^2}\right)_{x=x_n} = -0.0003$$

Velocity- First derivative
Acceleration- Second derivative

HOMEWORK

Q:1. The following data gives the velocity of a particle for 20 seconds at an interval of 5 seconds. Find the initial acceleration using the entire data.

Time t (sec):	0	5	10	15	20
Velocity v (m/sec):	0	3	14	69	228

ans:

t	v	Δv	$\Delta^2 v$	$\Delta^3 v$	$\Delta^4 v$
0	0				
		3			
5	3		8		
		11		36	
10	14		44		24
		55		60	
15	69		104		
		159			
20	228				

By Newton's forward interpolation formula, at $x = x_0$

$$\left(\frac{dv}{dt}\right)_{t=t_0} = \frac{1}{h}\left[\Delta v_0 + \frac{-1}{2}\Delta^2 v_0 + \frac{1}{3}\Delta^3 v_0 - \frac{1}{4}\Delta^4 v_0 + \cdots\right] \quad (h=5)$$

$$= \frac{1}{5}\left[3 - \frac{1}{2}\times 8 + \frac{1}{3}\times 36 - \frac{1}{4}\times 24 + \cdots\right]$$

$$\left(\frac{dv}{dt}\right)_{t=t_0} = \frac{1}{5}\left[3 - 4 + 12 - 6\right] = \frac{1}{5}[5] = 1$$

$\frac{d}{d}$ (Velocity) = acceleration.

Acceleration at $t=0 = \left(\frac{dv}{dt}\right)_{t=t_0} = 1 \text{ m/s}^2$

DERIVATIVE USING CENTRAL DIFFERENCE FORMULA (STIRLING'S FORMULA)

Consider Stirling's formula,

$$y(x) = y_0 + u\left[\frac{\Delta y_0 + \Delta y_{-1}}{2}\right] + \frac{u^2}{2!}\Delta^2 y_{-1} + \frac{u(u^2-1^2)}{3!}\left[\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right] + \frac{u^2(u^2-1^2)}{4!}\Delta^4 y_{-2} + \cdots\cdots$$

are

$$y(x) = y_0 + \frac{u}{2}[\Delta y_0 + \Delta y_{-1}] + \frac{u^2}{2}\Delta^2 y_{-1} + \frac{u^3-u}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{u^4-u^2}{24}\Delta^4 y_{-2} + \cdots \quad -(12)$$

where $u = \frac{x-x_0}{h}$.

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = \frac{1}{h}\left[\frac{1}{2}(\Delta y_0 + \Delta y_{-1}) + u\Delta^2 y_{-1} + \frac{(3u^2-1)}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{(u^3-2u)}{12}\Delta^4 y_{-2} + \cdots\right]$$

$$\frac{dy}{dx} = \frac{1}{h}\left[\frac{1}{2}(\Delta y_0 + \Delta y_{-1}) + u\Delta^2 y_{-1} + \frac{(3u^2-1)}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{2u^3-u}{12}\Delta^4 y_{-2} + \cdots\right] \quad -(13)$$

III^rly,

$$\frac{d^2y}{dx} = \frac{1}{h}\cdot\frac{1}{h}\left[\Delta^2 y_{-1} + \frac{6u}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{6u^2-1}{12}\Delta^4 y_{-2} + \cdots\right]$$

$$\frac{d^2y}{dx} = \frac{1}{h^2}\left[\Delta^2 y_{-1} + \frac{u}{2}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{6u^2-1}{12}\Delta^4 y_{-2} + \cdots\right]$$

III^rly,

$$\frac{d^3y}{dx^3} = \frac{1}{h^3}\left[\frac{1}{2}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{12u}{12}\Delta^4 y_{-2} + \cdots\right]$$

$$\Rightarrow \frac{d^3y}{dx^3} = \frac{1}{h^3}\left[\frac{1}{2}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + u\Delta^4 y_{-2} + \cdots\right] \quad -(14)$$

Let middle value be x_0.

Put $x = x_0$, i.e. $u = \frac{x - x_0}{h} = 0$.

Then eqns (13), (14), (15) becomes,

$$\left[\frac{dy}{dx}\right]_{x=x_0} = \frac{1}{h}\left[\frac{1}{2}(\Delta y_0 + \Delta y_{-1}) + \frac{-1}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \cdots\right]$$

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_0} = \frac{1}{h^2}\left[\Delta^2 y_{-1} - \frac{1}{12}\Delta^4 y_{-2} + \cdots\right]$$

$$\left[\frac{d^3y}{dx^3}\right]_{x=x_0} = \frac{1}{h^3}\left[\frac{1}{2}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \cdots\right]$$

central difference- if value is near to middle.

Q: Find the first and second derivative of the function at $x = 0.6$ from the below table.

x	0.4	0.5	0.6	0.7	0.8
y	1.5836	1.7974	2.0442	2.3275	2.6511

ans: Here, $x = x_0 = 0.6$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.4	1.5836				
		+0.2138			
0.5	1.7974		0.0330	$\Delta^3 y_{-2}$ 0.0035	
		Δy_{-1} 0.2468			$\Delta^4 y_{-2}$ 0.0003
0.6	y_0 2.0442		$\Delta^2 y_{-1}$ 0.0365		
		0.2833		0.0038	
0.7	2.3275	Δy_0	0.0403	$\Delta^3 y_{-1}$	
		0.3236			
0.8	2.6511				

At $x = x_0$, $h = 0.1$

$$\frac{dy}{dx} = \frac{1}{h}\left[\frac{1}{2}(\Delta y_0 + \Delta y_{-1}) + \frac{-1}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \cdots\right]$$

$$= \frac{1}{0.1}\left[\frac{1}{2}(0.2468 + 0.2833) - \frac{1}{12}(0.0035 + 0.0038) + \cdots\right]$$

$$\frac{dy}{dx} = \underline{2.644}$$

At $x=x_0$,

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_{-1} + \frac{-1}{12}\Delta^4 y_{-2} + \cdots\right]$$

$$= \frac{1}{(0.1)^2}\left[0.0365 - \frac{1}{12}(0.0003)\right]$$

$$\frac{d^2y}{dx^2} = 3.6475$$

are

$$\frac{dy}{dx} = 2.64442 \text{ and } \frac{d^2y}{dx^2} = 3.6475$$

At $x=x_0$,

$$\frac{dy}{dx} = \frac{1}{h}\left[\frac{1}{2}(\Delta y_0 + \Delta y_{-1}) - \frac{1}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{1}{60}(\Delta^5 y_{-2} + \Delta^5 y_{-3}) + \cdots\right]$$

If $x \neq x_0$, use general formulae (13), (14), (15).

$$\frac{dy}{dx} = \frac{1}{h}\left[\frac{1}{2}(\Delta y_0 + \Delta y_{-1}) + u\Delta^2 y_{-1} + \frac{(3u^2-1)}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{2u^3-u}{12}\Delta^4 y_{-2} + \cdots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_{-1} + \frac{u}{2}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{6u^2-1}{12}\Delta^4 y_{-2} + \cdots\right]$$

$$\frac{d^3y}{dx^3} = \frac{1}{h^3}\left[\frac{1}{2}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + u\Delta^4 y_{-2} + \cdots\right]$$

NUMERICAL INTEGRATION

The process of ~~evaluating~~ integrating a definite integral from the tabulated values of the integrand $f(x)$ is called Numerical Integration.

Given a set of $n+1$ paired values (x_i, y_i), where $0 \le i \le n$ (i varies from 0 to n), of the function $y=f(x)$ is not known explicitly, and it is required to compute $\int_{x_0}^{x_n} y\,dx$.

As we did in the case interpolation of numeric differentiation we replace $f(x)$ by an interpolating polynomial $P_n(x)$ - polynomial of x of degree n and obtain $\int_{x_0}^{x_n} P_n(x)\,dx$ which is approximately taken as the value for $\int_{x_0}^{x_n} f(x)\,dx$.

* (Not in Syllabus)

Newton - Cotes Quadrature Formula

($x_n = x_0 + nh$)

$$\int_{x_0}^{x_0+nh} f(x)\,dx = nh\left[y_0 + \frac{n}{2}\Delta y_0 + \frac{n(2n-3)}{12}\Delta^2 y_0 + \frac{n(n-2)^2}{24}\Delta^3 y_0 + \left(\frac{n^5}{5} - \frac{3n^3}{2} + \frac{11n^2}{3} - 3n\right)\frac{\Delta^4 y_0}{4!} + \cdots\right] - (1)$$

(3 marks)

TRAPIZOIDAL RULE

Put $n=1$ in the quadrature formula, interpolating polynomial will be a linear polynomial. $n=1$ only two data points.

$$\therefore \int_{x_0}^{x_0+h} f(x)\,dx \approx h\left[y_0 + \frac{1}{2}\Delta y_0\right]$$

$$\Delta y_0 = y_1 - y_0 \Rightarrow \int_{x_0}^{x_0+h} f(x)\,dx \approx h\left[y_0 + \frac{1}{2}(y_1 - y_0)\right]$$

$$\int_{x_0}^{x_0+h} f(x)\,dx \approx \frac{1}{2}h\left[y_0 + y_1\right]$$

Similarly, (next strip)

$$\int_{x_0+h}^{x_0+2h} f(x)\,dx = h\left[y_1 + \frac{1}{2}\Delta y_1\right]$$

$$\approx h\left[y_1 + \frac{1}{2}(y_2 - y_1)\right]$$

$$\int_{x_0+h}^{x_0+2h} f(x)\,dx \approx \frac{h}{2}[y_1 + y_2], \text{ etc...}$$

Continue similarly, we get,

$$\int_{x_0+(n-1)h}^{x_0+nh} f(x)\,dx \approx \frac{h}{2}[y_{n-1} + y_n]$$

Adding these 'n' integrals,

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{2}\left[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})\right]$$

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{2}\left[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})\right]$$

Q: Evaluate $\int_0^6 \frac{1}{(1+x^2)}\,dx$ by using trapizoid rule.

ans: Divide the interval 0 to 6 into 6 equal parts. (More no. of intervals - accuracy ↑).

∴ Here, $h = 1$ (0 to 1, 1 to 2, ...).

No. of intervals = 60

x	0	1	2	3	4	5	6
$f(x) = \frac{1}{1+x^2}$	1	0.5	0.2	0.1	0.0588	0.0385	0.0270
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{2}\left[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})\right]$$

(Trapizoidal rule).

$$\therefore \int_{x_0}^{6} \frac{1}{1+x^2}dx = \frac{1}{2}\left[(1+0.02870)+2(0.5+0.2+0.1+0.0588+0.0385)\right]$$

$$= \frac{1}{2}\left[1.02870 + 1.7946\right]$$

$$\int_{0}^{6} \frac{1}{1+x^2}dx = 1.410835$$

HOMEWORK.

Q: 1. Find the value of $f'(0.5)$ using Stirling's formula from the following data.

x:	0.35	0.40	0.45	0.50	0.55	0.60	0.65
$y=f(x)$:	1.521	1.506	1.488	1.467	1.444	1.418	1.389

ans:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$
0.35	1.521						
		-0.015					
0.40	1.506		-0.003				
		-0.018		0			
0.45	1.488		-0.003		0.001		
		-0.021		0.001		-0.003	
0.50	1.467		-0.002		-0.002		0.006
		-0.023		-0.001		0.003	
0.55	1.444		-0.003		0.001		
		-0.026		0			
0.60	1.418		-0.003				
		-0.029					
0.65	1.389						

$h = 0.05$

By stirling's formula, at $x = x_0$,

$$\frac{dy}{dx} = \frac{1}{h}\left[\frac{1}{2}(\Delta y_0 + \Delta y_{-1}) - \frac{1}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \frac{1}{60}(\Delta^5 y_{-2} + \Delta^5 y_{-3}) + \cdots\right]$$

$$= \frac{1}{0.05}\left[\frac{1}{2}(-0.023+0.021) - \frac{1}{12}(-0.001+0.001) + \frac{1}{60}(0.003+0.003)\right]$$

$$f'(0.5) = -0.438$$

Q:2. The table below gives the results of an observat
θ is the observed temperature in degrees centigrade
of a vessel of cooling water; t is the time
in minutes from the beginning of observation.

t:	1	3	5	7	9
θ:	85.3	74.5	67.0	60.5	54.3

a Find the approximate rate of cooling at t = 3 and 3.5,

ans:

t	θ	Δθ	Δ²θ	Δ³θ	Δ⁴θ
1	85.3				
		-10.8			
3	74.5		3.3		
		-7.50		-2.3	
5	67.0		1		1.6
		-6.50		-0.70	
7	60.5		0.30		
		-6.20			
9	54.3				

Rate of cooling $= \frac{d\theta}{dt}$

$u = \frac{t - t_0}{h} = \frac{t-1}{2}$ $\qquad h = 2$

At $t=3$, $u=1$.

At $t=3.5$, $u=1.25$.

Ft: $$\frac{d\theta}{dt} = \frac{1}{h}\left[\Delta\theta_0 + \frac{2u-1}{2}\Delta^2\theta_0 + \frac{3u^2-6u+2}{6}\Delta^3\theta_0 + \frac{(4u^3-18u^2+22u-6)}{24}\Delta^4\theta_0 + \cdots\right]$$

when $u=1$,

$$\frac{d\theta}{dt} = \frac{1}{h}\left[\Delta\theta_0 + \frac{1}{2}\Delta^2\theta_0 - \frac{1}{6}\Delta^3\theta_0 + \frac{1}{12}\Delta^4\theta_0 + \cdots\right]$$

$$= \frac{1}{2}\left[-10.8 + \frac{1}{2}(3.3) - \frac{1}{6}(-2.3) + \frac{1}{12}(1.6)\right]$$

$$\frac{d\theta}{dt} = -4.31667 //$$

Put $t=1.25$,

$$\frac{d\theta}{dt} = \frac{1}{h}\left[\Delta\theta_0 + 0.75\Delta^2\theta_0 + -0.1354 \times \Delta^3\theta_0 + (0.04948)\Delta^4\theta_0\right]$$

$$= \frac{1}{\frac{0.05}{2}}\left[-10.8 + (0.75 \times 3.3) - (0.1354 \times 2.3) + (0.4948 \times 1.6)\right]$$

$$\frac{d\theta}{dt} = -3.96718.$$

Q:3. Use the trapezoidal rule to evaluate $\int_{-3}^{3} x^4 dx$ and verify your results by actual integration.

ans. Here $y(x) = x^4$. Interval length $= (b-a) = 6$. So, we divide 6 equal intervals with $h = \frac{6}{6} = 1$.

x :	-3	-2	-1	0	1	2	3
y :	81	16	1	0	1	16	81

$$\int_{-3}^{3} y\,dx \approx \frac{h}{2}\left[(\text{sum of the first and last ordinates}) + 2(\text{sum of remaining ordinates})\right].$$

$$\int_{-3}^{3} y\,dx \approx \frac{1}{2}\left[(81+81) + 2(16+1+0+1+16)\right] \quad \rightarrow = [x^5]_0^3$$

$$\int_{-3}^{3} y\,dx = 115$$

Actual integration: $\int_{-3}^{3} x^4 dx = \frac{[x^5]_{-3}^{3}}{5} = \frac{2 \times 243}{5} = 97.2$

Q:4. Using trapezoidal rule evaluate the integral $\int_0^2 e^{x^2} dx$ by taking 10 intervals. $h = \frac{b-a}{h}$ (no. of intervals)

ans. $y(x) = e^{x^2}$ $\quad b-a = +2-0 = 2$

So we divide 10 equal parts $h = \frac{2}{10} = 0.2$.

x :	0	1	2	3	4	5	6	7	8	9
y :	1	2.718	54.598							

x :	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4
y :	1	1.0408	1.173	1.4333	1.8965	2.718	4.2207	7.099

x :	1.6	1.8	2.0
y :	12.936	25.534	54.598

$$\int_0^2 e^{x^2} dx = \frac{h}{2}\left[(\text{sum of first and last ordinates}) + 2(\text{sum of remaining ordinates})\right]$$

$$= \frac{0.2}{2}\left[(1+54.598) + 2(1.0408 + 1.173 + 1.4333 + 1.8965 + 2.718 + 4.2207 + 7.099 + 12.936 + 25.534)\right]$$

a

$$= 0.1\left[55.598 + 116.1026\right]$$

$$\int_0^2 e^{x^2} dx = 17.17006$$

04-10-2024

SIMPSON'S 1/3rd RULE

Putting $n=2$ in equation ① [Newton's Code Quadrature Formula),

$$\int_{x_0}^{x_0+2h} f(x)\,dx = 2h\left[y_0 + \frac{1}{4}\Delta y_0 + \frac{1}{6}\Delta^2 y_0\right]$$

$$= 2h\left[y_0 + (y_1 - y_0) + \frac{1}{6}(\Delta y_1 - \Delta y_0)\right]$$

$$= 2h\left[y_0 + (y_1 - y_0) + \frac{1}{6}\left[(y_2 - y_1) - (y_1 - y_0)\right]\right]$$

$$= 2h\left[y_0 + y_1 - y_0 + \frac{1}{6}\left[y_2 - 2y_1 + y_0\right]\right]$$

$$= 2h\left[y_1 - \frac{y_1}{3} + \frac{1}{6}y_2 + \frac{1}{6}y_0\right]$$

$$= 2h\left[\frac{2y_1}{3} + \frac{y_2}{6} + \frac{y_0}{6}\right]$$

$$= \frac{2h}{6}\left[\frac{4y_1 + y_2 + y_0}{6}\right]$$

$$= \frac{h}{3}\left[y_0 + 4y_1 + y_2\right]$$

$$\therefore \int_{x_0}^{x_0+2h} f(x)\,dx = \frac{h}{3}\left[y_0 + 4y_1 + y_2\right]$$

Next 3 points,

IIIrly, $$\int_{x_0+2h}^{x_0+4h} f(x)\,dx = \frac{h}{3}\left[y_2 + 4y_3 + y_4\right]$$

(no. of odd terms of y - x4 for $y_1, y_3 \ldots$ - twice for $y_2, y_4 \ldots$ - twice)

And so on.

$$\int_{x_0+(n-2)h}^{x_0+nh} f(x)\,dx = \frac{h}{3}\left[y_{n-2} + 4y_{n-1} + y_n\right]$$, n being even. (ex.)

Adding these integrals,

(y_0, y_n - do not repeat, $y_1, y_3, y_5, \ldots$ - x4, $y_2, y_4, \ldots$)

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{3}\left[(y_0+y_n) + 4(y_1+y_3+y_5+\ldots\ldots+y_{n-1}) + 2(y_2+y_4+\ldots\ldots+y_{n-2})\right]$$

NOTE

While applying Simpson's 1/3rd rule, the given interval must be divided into even no. of sub intervals.

Q: Evaluate $\int_0^6 \frac{1}{1+x^2}\,dx$ using simpson's 1/3rd rule.

ans: 0 - 6 - six sub intervals (EVEN).

x :	0	1	2	3	4	5	6
$f(x)=\frac{1}{1+x^2}$:	1	0.5	0.2	0.1	0.0588	0.0385	0.0270

By Simpson's 1/3rd rule,

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{3}\left[(y_0+y_n) + 4(y_1+y_3+y_5+\ldots\ldots+y_{n-1}) + 2(y_2+y_4+\ldots\ldots+y_{n-2})\right]$$

$h=1 \qquad n=6$

$$\int_0^6 \frac{1}{1+x^2}dx = \frac{1}{3}\left[(1+0.0270)+4(0.5+0.1+0.0385)+2(0.2+0.0588)\right]$$

$$= \frac{1}{3}\left[1.027+2.554+0.5176\right]$$

$$\int_0^6 \frac{1}{1+x^2}dx = 1.3662$$

c

Actual integration: $\tan^{-1}(x)_0^6 = \tan^{-1}(6) = 1.4056$ (radian)

6: Evaluate $\int_{-3}^{3} x^4 dx$ by using Simpson's 1/3rd rule.

ans: $n=6$ (six intervals) EVEN

x :	-3	-2	-1	0	1	2	3
$f(x)$:	81	16	1	0	1	16	81

By Simpson's 1/3rd rule, $h=1$

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{3}\left[(y_0+y_n)+4(y_1+y_3+y_5+\cdots+y_{n-1})+2(y_2+y_4+\cdots+y_{n-2})\right]$$

$$\int_{-3}^{3} x^4 dx = \frac{1}{3}\left[(81+81)+4(16+0+16)+2(1+1)\right]$$

$$\int_{-3}^{3} x^4 dx \approx 98$$

By Trapezoidal, $\int_{-3}^{3} x^4 dx \approx 115$

Actual integration, $\int_{-3}^{3} x^4 dx = 2\int_0^3 x^4 dx$

$$= \frac{2}{5}\left[x^4\right]_0^3 = 96.8$$

- More no·of intervals - more accurate answer·

(HOMEWORK) Problems

SIMPSON'S 3/8th RULE

- Put $n=3$ in Newton-Code's formula (eqn ①),

$$\int_{x_0}^{x_0+3h} f(x)dx = 3h\left[y_0+\frac{3}{2}\Delta y_0+\frac{3}{4}\Delta^2 y_0+\frac{1}{8}\Delta^3 y_0\right]$$

$\Delta y_2 - \Delta y_1 = y_3 - y_2 - y_2 + y_1$
$\downarrow$
$\Delta^2 y_1 - \Delta^2 y_0$

$$\int_{x_0}^{x_0+3h} f(x)\,dx = 3h\left[y_0+\frac{3}{2}(y_1-y_0)+\frac{3}{4}\underbrace{\left((y_2-y_1)-(y_1-y_0)\right)}_{\Delta y_1 \quad \Delta y_0}+\frac{1}{8}(\right.$$

$$\left.(y_3-y_2-y_2+y_1)-(y_2-2y_1+y_0)\right]$$

After simplification,

$$\int_{x_0}^{x_0+3h} f(x)dx = \frac{3h}{8}\left[y_0+3y_1+3y_2+y_3\right]$$

$$\int_{x_0+3h}^{x_0+6h} f(x)\,dx = \frac{3h}{8}\left[y_3+3y_4+3y_5+y_6\right]$$

(Interval must be multiple of 3)

Adding all such integrals from x_0 from x_0+nh, where n is a multiple of 3, we get,

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{3h}{8}\left[(y_0+y_n)+3(y_1+y_2+y_4+y_5+\ldots\ldots+y_{n-1})\right.$$
$$\left.+\ 2(y_3+y_6+\ldots\ldots+y_{n-3})\right]$$

↳ multiple of 3 terms = *2

NOTE

- While applying this rule, the no-of subintervals must be taken as multiple of 3.

Q: Evaluate $\int_0^6 \frac{1}{1+x^2}\,dx$ by using Simpson's 3/8 rule.

ans: No. of intervals = 6 (multiple of 3).

(Be careful, do not repeat y_n)

x	0	1	2	3	4	5	6
$f(x)=\frac{1}{1+x^2}$	1	0.5	0.2	0.1	0.0588	0.0385	0.027

By Simpson's 3/8 rule, $h=1$

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{3h}{8}\left[(y_0+y_n)+3(y_1+y_2+y_4+y_5+\ldots+y_{n-1})+2(y_3+y_6+\ldots+y_{n-3})\right]$$

$$\int_0^6 \frac{1}{1+x^2}\,dx = \frac{3}{8}\left[(1+0.027)+3(0.5+0.2+0.0588+0.0385)+2(0.1)\right]$$

$$\int_0^6 \frac{1}{1+x^2}\,dx = 1.3571$$

Trapezoidal = 1.4107 (more accurate)
↳ not always.

1/3 = 1.366

3/8 = 1.3571

Actual = 1.4056

Trapezoidal can be used in any condition:

10-2024

NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS

Only first order eqn

$$\frac{dy}{dx} = f(x,y) \qquad y(x_0) = y_0 \text{ (initial condition)}.$$

1. TAYLOR'S SERIES METHOD

Consider the first order equation,

$$\frac{dy}{dx} = f(x,y) \quad -①$$

given the initial condition $y(x_0)=y_0$ $-②$.

We expand $y(x)$ about the point $x=x_0$ in a Taylor series in powers of $x-x_0$.

$$y(x) = y(x_0) + \frac{(x-x_0)}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \ldots \quad -③$$

where $y^r(x_0) = \left(\frac{d^r y}{dx^r}\right)_{x=x_0}$

$$y(x) = y_0 + \frac{(x-x_0)}{1!} y_0' + \frac{(x-x_0)^2}{2!} y_0'' + \ldots\ldots\ldots\ldots$$

Diff b/t x value $= h$.

$x = x_1$,

$$y_1 = y_0 + \frac{h}{1!} y_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \cdots \quad (4)$$

where $h = x_1 - x_0$ (or $x_1 = x_0 + h$)

$y(x_0) = y_0$

$y'(x_0) = y_0'$ (given)

- To find y_0', y_0'', etc.....
- For finding y_0', y_0'', ... we use (1) and its derivate at $x = x_0$.
- Though the series (4) is an infinite series, we can truncate it at any ~~convinience~~ convinient term.
- If h is small and the desired accuracy is obtained.
- Having got y_1, we can find y_1', y_1'', y_1''', ... using $y' = f(x, y)$.
- Now expanding $y(x)$ ~~is a faulte~~ a Taylor series about the point $x = x_1$, we get,

$$y_2 = y_1 + \frac{h}{1!} y_1' + \frac{h^2}{2!} y_1'' + \frac{h^3}{3!} y_1''' + \cdots$$

~~$h = y_2$~~ $h = x_2 - x_1$ (constant).

- Proceeding the same manner we get,

$$y_{n+1} = y_n + \frac{h}{1!} y_n' + \frac{h^2}{2!} y_n'' + \frac{h^3}{3!} y_n''' + \cdots \quad (6)$$

This is the Taylor series for soln of ODE.

- ~~nth order algorithm~~ If we retain terms upto and including h^n and neglect terms involving h^{n+1} and higher powers of h, then Taylor algorithm is said to be n^{th} order.

eg If we retain upto $h^2 \Rightarrow 2^{nd}$ order.

Q: Solve $\frac{dy}{dx} = x+y$ given $y(1)=0$ and find $y(1.1)$ $y(x_1)$ and $y(1.2)$ $y(x_2)$ by Taylor's series method.

ans:

$\frac{dy}{dx} = x+y \Rightarrow y' = x+y$ $\quad y(1)=0$

$\therefore f(x,y) = x+y$ $\quad y(x_0)=y_0$

$\therefore x_0 = 1, \; y_0 = 0.$

$x_1 = 1.1 \quad y_1 = 1.2$

$h = 0.1 \quad (1.1-1.0)$

$y'' = 1+y' \qquad y''' = y''$

$y''' = y''$

$y' = x+y$

$\therefore y'_0 = (x_0+y_0) = 1+0 = 1$

$y''_0 = 1+y'_0 = 1+1 = 2 \qquad y'''_0 = y''_0 = 2$

$y''''_0 = y'''_0 = 2 \qquad y'''''_0 = y''''_0 = 2$

By Taylor series,

$$y_{n+1} = y_n + \frac{h}{1!}y'_n + \frac{h^2}{2!}y''_n + \frac{h^3}{3!}y'''_n + \frac{h^4}{4!}y''''_n + \frac{h^5}{5!}y'''''_n + \cdots$$

Put $n=0$,

$$y_1 = y_0 + h\,y'_0 + \frac{h^2}{2!}y''_0 + \frac{h^3}{3!}y'''_0 + \frac{h^4}{4!}y''''_0 + \frac{h^5}{5!}y'''''_0 + \cdots$$

$$y_1 = 0 + 0.1\times 1 + \frac{(0.1)^2}{2}\times 2 + \frac{(0.1)^3}{6}\times 2 + \frac{(0.1)^4}{24}\times 2 + \frac{(0.1)^5}{125}\times 2$$

$y_1 = 0.110341833 = y(1.1)$

$$y_2 = y_1 + \frac{h}{1!}y_1' + \frac{h^2}{2!}y_1'' + \frac{h^3}{3!}y_1''' + \frac{h^4}{4!}y_1'''' + \dots$$

$y' = x + y$

$y_1' = x_1 + y_1 = 1.1 + 0.1103341833 = 1.2103341833$

$y_1'' = 1 + y_1' = 1 + 1.2103341833 = 2.2103341833$

$y_1''' = y_1'' = 2.2103341833$

$y_1'''' = y_1''' = 2.2103341833$

$$y_2 = 0.1103341833 + \frac{(0.1)}{1} \times 1.2103341833 + \frac{(0.1)^2}{2} \times 2.2103341833$$

$$+ \frac{(0.1)^3}{6} \times 2.2103341833 + \frac{(0.1)^4}{24} \times 2.2103341833 + \dots$$

$y_2 = 0.24280160 = y(1.2)$

HOMEWORK

Q: Using Taylor's series method, find value of $y(0.1)$ correct to four decimal places, given $\frac{dy}{dx} = x^2 + y^2$, and $y(0) = 1$.

ans:

$\frac{dy}{dx} = x^2 + y^2$ $\qquad y(0) = 1$

$y' = x^2 + y^2$

$\therefore x_0 = 0 \qquad x_1 = 0.1 \qquad y_0 = 1$

$h = 0.1$

~~$y'' = 2x + y^2$~~ $\qquad y'' = 2x + 2yy'$

$y''' = 2 + 2(yy'' + (y')^2)$

$y''' = 2 + 2yy'' + 2(y')^2$

$$y''' = 2[1 + yy'' + (y')^2]$$

$$y'''' = 2[0 + yy''' + y'y'' + 2y'y''] = 2[yy''' + y'y'' + 2y'y'']$$

$$y''' = 2[yy''' + 3y'y'']$$

At $x = x_0$,

$$y_0' = x_0^2 + y_0^2 = 0 + 1 = 1$$

$$y_0'' = 2x_0 + 2y_0y_0' = 2\times0 + 2\times1\times1 = 2$$

$$y_0''' = 2[1 + y_0y_0'' + (y_0')^2] = 2[1 + 1\times2 + 1] = 8$$

$$y_0'''' = 2[y_0y_0''' + 3y_0'y_0''] = 2[1\times8 + 3\times1\times2]$$

$$y_0'''' = 28.$$

By Taylor's series,

$$y_{n+1} = y_n + \frac{h}{1!}y_n' + \frac{h^2}{2!}y_n'' + \frac{h^3}{3!}y_n''' + \frac{h^4}{4!}y_n''''$$

$$y_1 = y_0 + \frac{0.1}{1}\times y_0' + \frac{(0.1)^2}{2}\times y_0'' + \frac{(0.1)^3}{6}\times y_0''' + \frac{(0.1)^4}{24}\times y_0''''$$

$$= 1 + (0.1\times1) + \frac{(0.1^2\times2)}{2} + \frac{(0.1^3\times8)}{6} + \frac{(0.1^4\times28)}{24}$$

$$y_1 = 1.11145 = y(0.1)$$

HOMEWORK

Q:1. Evaluate the integral $I = \int_4^{5.2} \log_e x\,dx$ using Simpson's $\frac{1}{3}$ rule.

ans: $b - a = 5.2 - 4 = 1.2$.

Divide the interval into 6 equal parts (EVEN).

$$h = \frac{1.2}{6} = 0.2.$$

x :	4	4.2	4.4	4.6	4.8	5.0	5.2
$y = \log_e x$:	1.3862944	1.4350845	1.4816045	1.5260563	1.5686159	1.6094379	1.648658

~~$\log_x 2.303$~~

By Simpson's $\frac{1}{3}$rd rule,

$$\int_4^{5.2} \log_e x\,dx = \frac{h}{3}\left[(y_0+y_n)+2(y_2+y_4+\ldots)+4(y_1+y_3+\ldots)\right]$$

$$\int_4^{5.2} \log_e x\,dx = \frac{0.2}{3}\left[(1.386294+1.6486586)+2(1.4816045+1.5686159)+4(1.4350845+1.5260563+1.6094379)\right]$$

$$\int_4^{5.2} \log_e x\,dx = 1.82784724$$

Q:2. Use Simpson's $\frac{1}{3}$rd rule to find $\int_0^{0.6} e^{-x^2}dx$ by taking 7 ordinates. $(n=6)$

ans:

$$h = \frac{b-a}{n} = \frac{0.6-0}{6} = 0.1$$

x :	0.	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$:	1	1.0101	1.0408	1.0942	1.1735	1.2840	1.4333

By Simpson's $\frac{1}{3}$ rule,

$$\int_0^{0.6} e^{x^2}dx \approx \frac{h}{3}\left[(y_0+y_6)+4(y_1+y_3+y_5)+2(y_2+y_4)\right]$$

$$= \frac{0.1}{3}\left[(1+1.4333)+4(1.0101+1.0942+1.2840)+2(1.0408+1.1735)\right]$$

$$\approx 0.6805$$

$$\int_0^{0.6} e^{x^2}dx = 0.6805$$

Q:3. Evalcate $\int_{-3}^{3} x^4dx$ by using Simpson's $\frac{3}{8}$ rule.

ans: Divide into $n=6$ intervals (multiple of 3).

x :	-3	-2	-1	0	1	2	3
y :	81	16	1	0	1	16	81
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By Simpson's $\frac{3}{8}$th rule,

$$\int_{-3}^{3} y\,dx = \frac{3h}{8}\left[(y_0+y_6)+3(y_1+y_2+y_4+y_5)+2(y_3+y_6)\right]$$

$$\int_{-3}^{3} x^4 dx = \frac{3\times 1}{8}\left[(81+81)+3(16+1+1+16)+2(0)\right] = 99$$

Q: Evaluate the integral $\int_4^{5.2} \log_e x\,dx$ using Simpson's $\frac{3}{8}$th rule.

ans:

x	4	4.2	4.4	4.6	4.8	5.0	5.2
$f(x)$	1.3862944	1.4350845	1.4816045	1.5260563	1.5686159	1.6094379	1.6486586

$h = 0.2 = \frac{5.2-4}{6}$

By Simpson's $\frac{3}{8}$th rule,

$$\int_4^{5.2} y\,dx = \frac{3h}{8}\left[(y_0+y_6)+3(y_1+y_2+y_4+y_5)+2(y_3)\right]$$

$$= \frac{3\times 0.2}{8}\left[(1.3862944+1.6486586)+3(1.4350845+1.4816045+1.5686159+1.6094379)+2(1.5260563)\right]$$

$$\int_4^{5.2} \log_e x\,dx = 1.82784705$$

Q: 5 Evaluate $\int_0^6 \frac{1}{1+x}dx$ using (i) Trapezoidal rule (ii) Simpson's rule (both). Also check up by direct integration. (Taken $h=1$).

ans. $h = \frac{6-0}{6} = 1$. (Write formula)

x :	0	1	2	3	4	5	6
y :	1	0.5	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$.

(i) By Trapezoidal rule: $h=1$

$$\int_0^6 \frac{dx}{1+x} = \frac{1}{2}\left[(1+\tfrac{1}{7})+2(\tfrac{1}{3}+\tfrac{1}{4}+\tfrac{1}{5}+\tfrac{1}{6})\right]$$

$$\int_0^6 \frac{dx}{1+x} = 2.02142857$$

(ii) By Simpson's 1/3rd rule,

$$\int_0^6 \frac{1}{1+x}\,dx = \frac{1}{3}\left[\left(1+\frac{1}{7}\right)+2\left(\frac{1}{3}+\frac{1}{5}\right)+3\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{6}\right)\right]$$

$$\int_0^6 \frac{1}{1+x}\,dx = 1.958730159$$

By Simpson's 3/8th rule,

$$\int_0^6 \frac{1}{1+x}\,dx = \frac{3}{8}\left[\left(1+\frac{1}{7}\right)+3\left(0.5+\frac{1}{3}+\frac{1}{5}+\frac{1}{6}\right)+2\left(\frac{1}{4}\right)\right]$$

$$\int_0^6 \frac{1}{1+x}\,dx = 1.966071429$$

Actual integration,

$$\int_0^6 \frac{1}{1+x}\,dx = \left[\log_e(1+x)\right]_0^6 = 1.94591015$$

08-08-2024

a: Using Taylor series method, compute $y(0.2)$ and $y(0.4)$, correct to 4 decimal places given $\frac{dy}{dx} = 1-2xy$ and $y(0)=0$.

ans:

$y' = 1-2xy$ $\quad y(0)=0$

$x_0 = 0$ $\quad y_0 = 0$ $\quad x_1 = 0.2$ $\quad x_2 = 0.4$

$h = 0.2$

~~$y'' = -2y'$~~ $\quad$ ~~$y''' = -2$~~

$y'' = -2(xy' + y^*)$

$y''' = -2(xy'' + y' + y') = -2(xy'' + 2y')$

$y'''' = -2(xy''' + y'' + 2y'') = -2(xy''' + 3y'')$

~~At $x=x_0$~~, $y''''' = -2(xy'''' + y''' + 3y''') = -2(xy'''' + 4y''')$

By Taylor series method,

$$y_{n+1} = y_n + \frac{h}{1!}y_n' + \frac{h^2}{2!}y_n'' + \frac{h^3}{3!}y_n''' + \frac{h^4}{4!}y_n'''' + \dots$$

$y_0' = 1$ $\quad y_0'' = 0$ $\quad y_0''' = -4$ $\quad y_0'''' = 0$

$y_0''''' = 32$

$$y_1 = y_0 + (0.2 \times 1) + \frac{(0.2^2 \times 0)}{2} + \frac{(0.2^3 \times -4)}{6} + \frac{(0.2^4 \times 0)}{24} + \frac{(0.2^5 \times 32)}{120}$$

$y_1 = 0.194752$

$y_1' = 1 - 2(0.2 \times 0.194752) = 0.9220992$

$y_1'' = -2(0.2 \times 0.9220992 + 0.194752) = -0.75834368$

$y_1''' = -2(0.2 \times -0.75834368 + 0.9220992 \times 2) = -3.385059328$

$y_1'''' = -2(0.2 \times -3.385059328 + 3 \times -0.75834368) = 5.904085811$

$y_1''''' = -2(0.2 \times 5.904085811 + 4 \times -3.385059328) = 24.71884403$

$$y_2 = y_1 + (0.2 \times y_1') + \frac{(0.2)^2}{2!} \times y_1'' + \frac{(0.2)^3}{3!} \times y_1''' + \frac{(0.2)^4}{4!} \times y_1'''' + \frac{(0.2)^5}{5!} \times y_1'''''$$

$$y_2 = 0.194752 + (0.2 \times 0.9220992) + \frac{(0.2^2 \times -0.75834368)}{2} + \frac{(0.2^3 \times -3.385059328)}{6} + \frac{(0.2^4 \times 5.904085811)}{24} + \frac{(0.2^5 \times 24.7188403)}{120}$$

$$y_2 = 0.3599511$$

EULER'S METHOD

(similar to actual answer).

$$y_{n+1} = y_n + hf(x_n, y_n) \quad ; \quad n = 0, 1, \ldots$$

$$y(x+h) = y(x) + hf(x, y)$$

When $n=0$, $y_1 = y_0 + hf(x_0, y_0)$

$\frac{dy}{dx} = f(x, y)$ & $y(x_0) = y_0$

$y' = f(x, y) \Rightarrow y_0' = f(x_0, y_0)$

$\therefore y_1 = y_0 + hy_0'$

$y_2 = y_1 + hy_1'$

Q: Given $y' = -y$ and $y(0) = 1$, determine the values of y at $x = 0.01, 0.02, 0.03, 0.04$.

ans: $y' = -y$

$f(x, y) = -y$ $\quad y(x_0) = y_0$ $\quad y(0) = 1$ $\quad x_0 = 0, \; y_0 = 1$

c

$x_1 = 0.01$

$x_2 = 0.02$

$x_3 = 0.03$

$x_4 = 0.04$

$h = x_2 - x_1 = 0.01$

By Euler's formula,

$$y_{n+1} = y_n + hf(x, y)$$

$$y_1 = y_0 + hy_0'$$

$\Rightarrow y_1 = 1 + 0.01 \times -1 \quad (y_0' = -y_0 = -1)$

$y_1 = 0.99$

$$y_2 = y_1 + hy_1' \qquad (y_1' = -y_1 = -0.99)$$

$y_2 = 0.99 + (0.01 \times -0.99)$

$y_2 = 0.9801$

$y_3 = y_2 + hy_2'$

$y_3 = 0.9801 + (0.01 \times -0.9801) = 0.970299$

$y_4 = y_3 + hy_3'$

$y_4 = 0.970299 + (0.01 \times -0.970299) = 0.960596$

$y(0.01) = 0.99$

$y(0.02) = 0.9801$

$y(0.03) = 0.970299$

$y(0.04) = 0.960596$

Actual method:

$$\frac{dy}{dx} = -y$$

$$-\frac{dy}{y} = dx$$

$\log y = x$

$y = e^{-x}$

$y = 0.99 \quad (x =$

$y = 0.9802 \quad (x$

$y = 0.9704 \quad (x = 0$

$y = 0.960789 \quad (x$

Q: Solve using Euler's method, $y' = x+y$, $y(0)=1$.
For $x = 0.2, 0.4, 0.6, 0.8, 1$. Check the answer with exact solution.

Ans:

$y' = x+y$

$h = 0.2$ $\quad x_0 = 0$ $\quad y_0 = 1$

$x_1 = 0.2$ $\quad x_3 = 0.6$ $\quad x_5 = 1$
$x_2 = 0.4$ $\quad x_4 = 0.8$

$y_{n+1} = y_n + h y_n'$

$y_1 = y_0 + hy_0' = 1 + 0.2(0+1) = 1.2$ $\quad (y_0' = x_0 + y_0 = 0+1)$

$y_2 = y_1 + hy_1' = 1.2 + 0.2(0.2+1.2) = 1.48$

$y_3 = y_2 + hy_2' = 1.48 + 0.2(0.4+1.48) = 1.8856$

~~$y_4 = y_3 + hy_3' = 1.5552 + 0.2(0.1.5552+0.6) = 1.986$~~

$y_4 = y_3 + hy_3' = 1.856 + 0.2(0.6+1.856) = 2.3472$

$y_5 = y_4 + hy_4' = 2.3472 + 0.2(0.8+2.3472) = 2.97664$

Exact solution,

Ans: $y = 2e^x - x - 1$

$\frac{dy}{dx} = x+y$

Let $x+y = t$

$\frac{dt}{dx} = 1 + \frac{dy}{dx}$

$\frac{dy}{dx} = \frac{dt}{dx} - 1$

$\frac{dt}{dx} - 1 = x+y$ ~~t~~

$\Rightarrow \frac{dt}{dx} = t+1 \Rightarrow \frac{dt}{t+1} = dx$

Integrating $\Rightarrow \ln(t+1) = x+C \Rightarrow x+y = e^{x+C} - 1$

$y(0) = 1 \Rightarrow 0+1 = e^C - 1 \Rightarrow e^C = 2$

$\therefore$ ~~$\ln($~~ $y = 2e^x - x - 1$

MODIFIED EULER'S METHOD

$$y_{n+1} = y_n + h\left[f\left(x_n + \frac{1}{2}h,\ y_n + \frac{1}{2}h f(x_n, y_n)\right)\right]$$

or

$$y(x+h) = y(x) + h\left[f\left(x + \frac{1}{2}h,\ y + \frac{1}{2}h f(x,y)\right)\right]$$

don't forget $f(\ )$

Q: Compute y at $x = 0.25$ by modified Euler method given $y' = 2xy$ and $y(0) = 1$.

ans: $y' = f(x,y)$ $\quad f(x,y) = 2xy$ $\quad x_0 = 0$ $\quad y_0 = 1$

$x_1 = 0.25$

$h = x_1 - x_0 = 0.25$.

$$y_{n+1} = y_n + h\left[f\left(x_n + \frac{1}{2}h,\ y_n + \frac{1}{2}h f(x_n, y_n)\right)\right]$$

$$y_1 = y_0 + 0.25\left[f\left(x_0 + \frac{1}{2}0.25,\ y_0 + \frac{1}{2}h f(x_0, y_0)\right)\right]$$

$x_0 + \frac{1}{2}0.25 = 0.125$

$y_0 + \frac{1}{2}h f(x_0, y_0) = 1 + \frac{0.25}{2} \times 0 = 1$

$y_1 = 1 + 0.25\ [f(0.125, 1)]$

$y_1 = 1 + 0.25(2 \times 0.125 \times 1) = 1.0625$

$y(0.25) = 1.0625$

Actual: $\quad \frac{dy}{dx} = 2xy$

$\frac{dy}{y} = 2x\,dx$

Sing $\Rightarrow$ $\log y = x^2$

$y = e^{x^2} = 1.06449$

H.W.

Q: solve the equation $\frac{dy}{dx} = 1-y$ given $y(0)=0$ using modified Euler's method and calculate the solutions at $x = 0.1, 0.2, 0.3$. Compare the results with exact solution.

ans:

$y' = 1-y$ $\quad f(x,y) = 1-y$ $\quad x_0 = 0$ $\quad y_0 = 0$

$h = 0.1$ $\quad x_1 = 0.1$ $\quad x_2 = 0.2$ $\quad x_3 = 0.3$

By modified Euler's method,

$$y_{n+1} = y_n + h\left[f\left(x_n + \frac{1}{2}h,\ y_n + \frac{1}{2}hf(x_n, y_n)\right)\right]$$

$$y_1 = y_0 + h\left[f\left(x_0 + \frac{1}{2}h,\ y_0 + \frac{1}{2}hf(x_0, y_0)\right)\right]$$

$$y_1 = 0 + 0.1\left[f(0.05, 0.05)\right] = 0.095$$

$$y_2 = y_1 + h\left[f\left(x_1 + \frac{1}{2}h,\ y_1 + \frac{1}{2}hf(x_1, y_1)\right)\right]$$

$$y_2 = 0.095 + 0.1\left[f(0.145, 0.14025)\right] = 0.180975$$

$$y_3 = y_2 + h\left[f\left(x_2 + \frac{1}{2}h,\ y_2 + \frac{1}{2}hf(x_2, y_2)\right)\right]$$

$$y_3 = 0.180975 + 0.1\left[f\left(0.2 + \frac{0.1}{2},\ 0.180975 + \frac{1}{2}\times 0.1 f(0.2, 0.180975)\right)\right]$$

$$= 0.180975 + 0.1\left[f(0.25, 0.22192625)\right]$$

$$y_3 = 0.25878\,2375$$

Exact soln

$\left(\frac{dy}{dx} + Py = Q\right)$

$\frac{dy}{dx} = 1-y \Rightarrow \frac{dy}{1-y} = dx \Rightarrow \frac{dy}{dx} + y = 1$

$P = 1 \quad Q = 1$

$e^{\int P dx} = e^x$

Soln: $y \times e^x = \int e^x dx$

$y(0) = 0$

$1 = e^c$

$y = e^x - e^x + 1 = 1 - e^x$

Integrating $\Rightarrow \log(1-y) = x + C$

$\Rightarrow 1 - y = e^x e^C \Rightarrow y = 1 + e^x + C$

$y e^x = e^x + C$

$x_1 = 0.1, \quad y(0) = 0, \quad C = -1,$

$\therefore y e^x = e^x - 1$

$x = 0.1, \quad y = 0.0952$

$x = 0.2, \quad y = 0.18127$

$x = 0.3, \quad y = 0.259182$ //

RUNGE-KUTTA METHOD (RK Method)

- Not mentioned do 4th order of Runge-Kutta method.
- Drawback of Taylor series: need to find higher order deriva

2nd order Runge-Kutta Method (for first order ODE)

$$k_1 = h f(x, y)$$

$$k_2 = h f\left(x + \frac{1}{2}h,\ y + \frac{k_1}{2}\right)$$

where $\Delta y = k_2$ and $h = \Delta x$.

$$y(x+h) = y(x) + \Delta y$$

Fourth order Runge-Kutta Method (for first order ODE)

(Default method)

(Be careful with formula)

$$k_1 = hf(x, y)$$

$$k_2 = hf\left(x + \frac{1}{2}h,\ y + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x + \frac{1}{2}h,\ y + \frac{k_2}{2}\right)$$

$$k_4 = hf(x + h,\ y + k_3)$$

(no $\frac{1}{2}$ here)

$$\Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$y(x+h) = y(x) + \Delta y.$$

$$y_1 = y_0 + \Delta y_0$$

REMARK

For second order RK method,

$y_1 = y_0 + \Delta y_0$ $\quad \Delta y_0 = k_2 = hf\left(x_0 + \frac{1}{2}h,\ y_0 + \frac{1}{2}hf(x_0, y_0)\right)$

$\Delta y_0 = k_2$, $\quad y_1 = y_0 + k_2 = y_0 + hf\left(x_0 + \frac{1}{2}h,\ y_0 + \frac{k_1}{2}\right)$

$$y_1 = y_0 + hf\left(x_0 + \frac{1}{2}h,\ y_0 + \frac{1}{2}hf(x_0, y_0)\right)$$

$=$

↓

This is modified Euler method.

∴ Second order RK method is modified Euler method.

Q: Apply 4th order RK method to find $y(0.2)$ given that $y' = x+y$, $y(0)=1$. [Take $h=0.1$].

Ans:

$x_0 = 0$ $\quad y_0 = 1$ $\quad h = 0.1$

$f(x,y) = x+y$ $\quad \therefore x_1 = 0.1$, $x_2 = 0.2$

By fourth order RK method.

$$k_1 = hf(x,y)$$

For y_1, $x_1 = 0.1$,

$$k_1 = 0.1\ f(x_0,y_0) = 0.1 \times (0+1) = 0.1$$

$$k_2 = hf\left(x_0+\frac{1}{2}h,\ y_0+\frac{k_1}{2}\right)$$

$$= hf\left(\frac{0.1}{2},\ 1+\frac{0.1}{2}\right) = 0.1 \times f(0.05,\ 1.05)$$

$$k_2 = 0.1(0.05+1.05) = 0.11$$

$$k_3 = hf\left(x_0+\frac{1}{2}h,\ y_0+\frac{k_2}{2}\right)$$

$$= 0.1\ f\left(0+\frac{0.1}{2},\ 1+\frac{0.11}{2}\right) = 0.1\ f(0.05,\ 1.055)$$

$$k_3 = 0.1105$$

$$k_4 = hf(x_0+h,\ y_0+k_3) = 0.1\ f(0.1,\ 1.1105)$$

$$k_4 = 0.12105$$

$$\Delta y = \frac{1}{6}(k_1+2k_2+2k_3+k_4)$$

$$= \frac{1}{6}(0.1+(2\times 0.11)+(2\times 0.1105)+0.12105)$$

$$\Delta y_0 = 0.11034$$

$$y(x+h) = y(x)+\Delta y$$

$$y_1 = y_0+\Delta y_0 = 1+0.11342 = 1.11034\,2$$

$$y(0.1) = 1.110342$$

For y_2, $x_2 = 0.2$,

($x_1 = 0.1$, $y_1 = 1.110342$)

$k_1 = hf(x_1, y_1) = 0.1\ f(0.1, 1.110342)$

$k_1 = 0.121034$

$k_2 = hf\left(x_1 + \frac{1}{2}h, y_1 + \frac{k_1}{2}\right)$

$= 0.1\ f\left(0.1 + \frac{0.1}{2}, 1.110342 + \frac{0.121034}{2}\right)$

$= 0.1\ f(0.15, 1.170859)$

$k_2 = 0.13208 59$

$k_3 = hf\left(x_1 + \frac{1}{2}h, y_1 + \frac{k_2}{2}\right)$

$= 0.1\ f\left(0.1 + \frac{0.1}{2}, 1.110342 + \frac{0.1320859}{2}\right)$

$= 0.1\ f(0.15, 1.17638495)$

$k_3 = 0.132638495$

$k_4 = hf(x_1 + h, y_1 + k_3)$

$= 0.1\ f(0.1 + 0.1, 1.110342 + 0.132638495)$

$= 0.1\ f(0.2, 1.242980495)$

$k_4 = 0.144298049$

$\Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$\Delta y_1 = \frac{1}{6}(0.121034 + (2 \times 0.1320859) + (2 \times 0.132638495) + 0.144298049)$

$\Delta y_1 = 0.132463473$

$y_2 = y_1 + \Delta y_1$

$y_2 = 1.110342 + 0.132463473$

$y_2 = 1.242805473$

$\therefore y(0.2) = 1.2428054\overline{73}$

H.W

Q: Obtain the values of y at $x = 0.1, 0.2$ using RK method of (i) 2nd order (ii) 4th order for the differential equation $y' = -y$ given $y(0) = 1$.

Ans: $f(x, y) = -y$ $\quad x_0 = 0$ $\quad y_0 = 1$

$x_1 = 0.1$ $\quad x_2 = 0.2$

$h = 0.1$

(i) 2nd order:

$k_1 = h f(x_0, y_0) = (0.1) f(0, 1) = 0.1 \times -1 = -0.1$

$k_2 = h f\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) = 0.1 f(0.05, 0.95)$

$= -0.1 \times 0.95 = -0.095 = \Delta y$

$y_1 = y_0 + \Delta y = 1 - 0.095 = 0.905$

$y_1 = y(0.1) = 0.905$

For y_2, $(x_1, y_1) = (0.1, 0.905)$

$k_1 = 0.1 f(x_1, y_1) = 0.1 \times -0.905 = -0.0905$

$k_2 = h f\left(x_1 + \frac{1}{2}h, y_1 + \frac{1}{2}k_1\right)$

$k_2 = 0.1 f(0.15, 0.85975) = -0.085975$

$\Delta y = k_2$

$y_2 = y(0.2) = y_1 + \Delta y = 0.819025$

(ii) 4th order.

$k_1 = hf(x_0, y_0) = (0.1) f(0, 1) = -0.1$

$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) = (0.1) f(0.05, 0.95) = -0.095$

$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right) = (0.1) f(0.05, 0.9525) = -0.09525$

$k_4 = hf(x_0 + h, y_0 + k_3) = (0.1) \times f(0.1, 0.90475) = -0.090475$

$$\Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$y_1 = y_0 + \Delta y = 1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$y_1 = y(0.1) = 0.9048375$$

Again $(x_1, y_1) = (0.1,\ 0.9048375)$.

For y_2,

$$k_1 = hf(x_1, y_1) = (0.1)x - y_1 = -0.09048375$$

$$k_2 = hf\left(x_1 + \frac{1}{2}h,\ y_1 + \frac{1}{2}k_1\right)$$

$$k_2 = (0.1)\ f(0.15,\ 0.8595956) = -0.08595956$$

$$k_3 = hf\left(x_1 + \frac{1}{2}h,\ y_1 + \frac{1}{2}k_2\right)$$

$$k_3 = (0.1)f(0.15,\ 0.8618577) = -0.08618577$$

$$k_4 = hf(x_1 + h,\ y_1 + k_3)$$

$$k_4 = (0.1)f(0.2,\ 0.8186517) = -0.08186517$$

~~$\Delta y = \frac{1}{6}(y_1 + 2y_2 + 2y$~~

$$\Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}(-0.09048375 - (2\times 0.08595956) - (2\times 0.08618577) - 0.08186517)$$

$$\Delta y = -0.086106607$$

$$y_2 = y(0.2) = y_1 + \Delta y = 0.81873089$$

EULER'S METHOD

We have to solve $\frac{dy}{dx} = f(x,y)$ with initial condition,

$$y(x_0) = y_0 \quad -(1)$$

Let us take the points $x = x_0, x_1, x_2, \ldots\ldots$ where $x_i - x_{i-1} = h$.

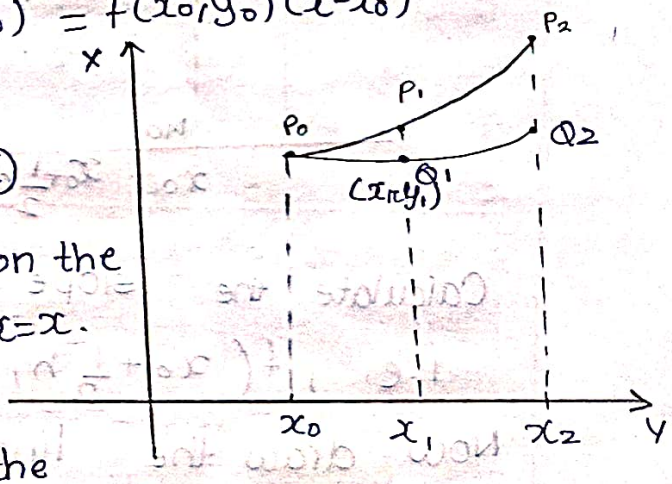
i.e., $x_i = x_0 + ih$, $i = 0, 1, 2, \ldots\ldots$

Let the actual solution of the differential ~~so~~ equation be denoted by the graph. $P_0(x_0, y_0)$ lies on the curve. We require the value of y of the curve at $x = x_i$

The equation of the tangent at (x_0, y_0) to the curve is,

$$y - y_0 = y'_{(x_0,y_0)}(x - x_0) = f(x_0,y_0)(x - x_0)$$

$$y = y_0 + f(x_0,y_0)(x - x_0) \quad -(2)$$



This y is the value of y on the tangent corresponding to $x = x$.

In the interval (x_0, x_1) the curve is approximated by the tangent.

∴ The value of y on the curve is approximately equal to the value of y on the tangent at (x_0, y_0) corresponding to $x = x_1$.

$$y_1 = y_0 + f(x_0,y_0)(x_1 - x_0)$$

$$y_1 = y_0 + hy'_0 \quad \text{where} \quad h = x_1 - x_0.$$

Again, we approximate curve by the line through (x_1, y_1) and whose slope is $f(x_1, y_1)$, we get,

$$y_2 = y_1 + hf(x_1,y_1) = y_1 + hy'_1$$

Thus, $y_{n+1} = y_n + hf(x_n, y_n)$, $n = 0, 1, 2, \ldots$

This is called Euler's algorithm.

In other words, $y(x+h) = y(x) + hf(x, y)$.

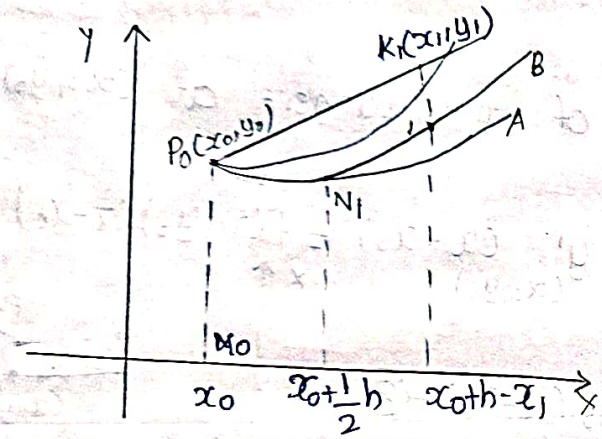
Modified Euler Method

Let $P_0(x_0, y_0)$ be the point on the solution curve.

Let P_0A be the tangent at (x_0, y_0) to the curve.

Now let this tangent meet the ordinate at $x = x_0 + \frac{1}{2}h$ at N_1.

y-coordinate of $N_1 = y_0 + \frac{1}{2}hf(x_0, y_0)$ — ①



Calculate the slope at N_1.

i.e., $f\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}hf(x_0, y_0)\right)$.

Now draw the line through $P(x_0, y_0)$ with the slope as the slope. Let this line meet $x = x_1$ at $K_1(x_1, y_1^{(1)})$.

This $y_1^{(1)}$ is taken as the approximate value of y at $x = x_1$.

$\therefore y_1^{(1)} = y_0 + h\left[f\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}hf(x_0, y_0)\right)\right]$

In general,

$$y_{n+1} = y_n + h\left[f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}hf(x_n, y_n)\right)\right] \quad -②$$

or

$$y(x+h) = y(x) + h\left[f\left(x + \frac{1}{2}h, y + \frac{1}{2}hf(x, y)\right)\right] \quad -③$$

and (3) are called modified Euler's formula.

NOTE

- There is a lot of confusion among the authors. Some take the improved Euler's method as the modified Euler method and the modified Euler method is not mentioned at all. ~~to~~